

Minimal Surfaces and the Plateau Problem

Mentor: Gustavo Martins

Josh Johnson

October 2026

Overview

This material is adapted from Chapters 1 and 4 of “A Course in Minimal Surfaces” by Colding and Minicozzi.

- Definition and examples
- Plateau's problem
- Sketch of the proof

Definition

- A *minimal surface* is a critical point of the area functional.
- For open bounded $\Omega \subset \mathbb{R}^2$ and $u \in C^2(\Omega)$,

$$\text{Area}(\text{Graph}_u) = \int_{\Omega} \sqrt{1 + |\nabla u|^2}.$$

- Perturb u by $\eta \in C^2(\Omega)$ such that $\eta|_{\partial\Omega} = 0$. Why?
 - Drop the boundary term in the divergence theorem.
 - Fixes the boundary data and hints towards the Plateau problem.

Definition

- Then

$$\begin{aligned}\left. \frac{d}{dt} \right|_{t=0} \text{Area}(\text{Graph}_{u+t\eta}) &= \int_{\Omega} \frac{\langle \nabla u, \nabla \eta \rangle}{\sqrt{1 + |\nabla u|^2}} \\ &= - \int_{\Omega} \eta \operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right).\end{aligned}$$

- The *minimal surface equation for graphs*:

$$\operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) \equiv 0.$$

- For graphs, this is a minimum for the area functional.

Definition

- More generally, on a Riemannian manifold (M, g) with covariant derivative ∇ , we can take a k -dimensional submanifold $\Sigma \subset M$.
- We get an induced covariant derivative ∇_Σ on Σ and second fundamental form

$$A(X, Y) = (\nabla_X Y)^N$$

for $X, Y \in T_x \Sigma$.

Definition

- The mean curvature at $x \in \Sigma$ is

$$H = \sum_{i=1}^k A(E_i, E_i),$$

where E_i is an orthonormal basis for $T_x \Sigma$.

- Σ is a critical point of the area functional iff $H \equiv 0$.

Examples

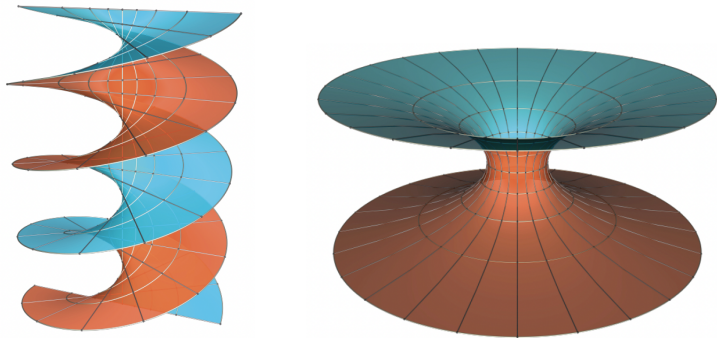


Figure: Left: helicoid, surface formed by soap film when dipping a wire frame in solution. Right: catenoid, surface of revolution of hyperbolic cosine. (Images taken from Colding, Minicozzi)

Examples

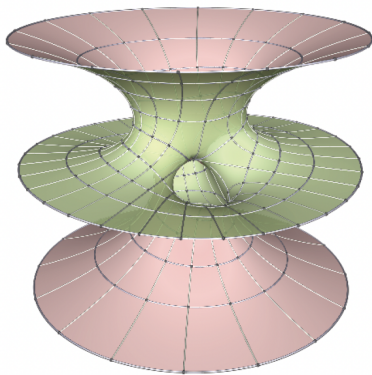


Figure: Costa-Hoffman-Meeks surface: finite total curvature, genus one, and three embedded ends. (Image taken from Colding, Minicozzi)

Overview

- Definition and examples
- Plateau's problem
- Sketch of the proof

Plateau Problem

- Given a closed curve $\Gamma \subset \mathbb{R}^3$, can we find a minimal surface with boundary Γ ?
- This problem was worked on by Lagrange, Plateau, Lebesgue, and others.
- Solved by Douglas and Rado for functions on the disk $D \subset \mathbb{R}^2$.

Plateau Problem

Theorem: Given a piecewise C^1 closed Jordan curve $\Gamma \subset \mathbb{R}^3$, define

$$X_\Gamma = \{\psi : D \rightarrow \mathbb{R}^3 : \psi \in C^0(\bar{D}) \cap W^{1,2}(D) \\ \text{and } \psi|_{\partial D} : \partial D \rightarrow \Gamma \text{ is monotone and surjective}\}.$$

Then there is a map $u \in X_\Gamma \cap C^\infty(D)$ such that the image of u minimizes area among all maps from the disk with boundary Γ .

Overview

- Definition and examples
- Plateau's problem
- Sketch of the proof

From here on, things get a bit technical.

Key Results

- For $u \in W^{1,2}(D)$, we define the energy of u :

$$E(u) = \frac{1}{2} \int_D |\nabla u|^2.$$

- One can show that

$$\inf_{\psi \in X_\Gamma} \text{Area}(\psi) = \inf_{\psi \in X_\Gamma} E(\psi) \equiv E_\Gamma.$$

Key Results

- **Dirichlet Problem:** If $w \in C^0(\overline{D}) \cap W^{1,2}(D)$, then there exists a unique, energy-minimizing harmonic function $u \in C^0(\overline{D}) \cap W^{1,2}(D) \cap C^\infty(D)$ such that $u = w$ on ∂D .
- **Rellich Compactness:** If $u_k \in W^{1,2}(D)$ is such that $\sup_k \|u_k\|_{W^{1,2}} < \infty$, then there is a subsequence $u_{k'}$ and $u \in W^{1,2}(D)$ such that $u_{k'} \rightarrow u$ strongly in L^2 , $\nabla u_{k'} \rightarrow \nabla u$ weakly in L^2 , and

$$E(u) \leq \liminf E(u_k).$$

Key Results

- Given $p \in D$, for $\rho > 0$ define

$$C_\rho = \{q \in D : |p - q| = \rho\}.$$

- For $u : D \rightarrow \mathbb{R}^3$, let $d(C_\rho)$ be the diameter of $u(C_\rho)$.
- Courant-Lebesgue lemma:** if $u \in C^0(\overline{D}) \cap W^{1,2}(D)$ and $E(u) \leq K/2$, then for every $\delta < 1$, there is $\rho \in [\delta, \sqrt{\delta}]$ such that

$$(d(C_\rho))^2 \leq \frac{8\pi^2 K}{-\log \delta}.$$

Key Results

- A diffeomorphism $\phi : D \rightarrow D$ is *conformal* if it preserve angles. Energy is conformally invariant, i.e., $E(u) = E(u \circ \phi)$.
- The conformal group of D is that of linear fractional transformations which acts triply transitively on the boundary.
- Fix $p_i \in \partial D$ and $q_i \in \Gamma$ for $i = 1, 2, 3$.
- **Lemma:** For any $K > 0$, let $\mathcal{F}$ be the family of $\psi \in X_\Gamma$ with $E(\psi) \leq K/2$ and $\psi(p_i) = q_i$. Then $\mathcal{F}$ is equicontinuous on ∂D .

Proof Sketch

- First, show that $E_\Gamma < \infty$.
- Next, take a sequence $u_j \in X_\Gamma$ with $E(u_j) \rightarrow E_\Gamma$. By the Dirichlet problem and composing with linear fractional transformations, we can choose u_j to be harmonic and in $\mathcal{F}$.
- By Rellich compactness, there is a weakly convergent subsequence $u_k \rightarrow u \in W^{1,2}$ with $E(u) \leq E_\Gamma$.

Proof Sketch

- By the lemma, $\{u_j|_{\partial D}\}$ is equicontinuous on ∂D , so by Arzela-Ascoli, we extract a uniformly convergent subsequence.
- Moreover, since each u_j is harmonic, so is $u_j - u_k$, so the maximum principle gives

$$\sup_D |u_j - u_k| = \max_{\partial D} |u_j - u_k|,$$

which implies uniform convergence on $\overline{D}$.

- Thus, $u \in X_\Gamma \cap C^\infty(D)$ and $E(u) = E_\Gamma$.

Thanks for listening!

References:

- T. H. Colding and W. P. Minicozzi, *A Course on Minimal Surfaces*
- M. P. do Carmo, *Differential Geometry of Curves and Surfaces*
- M. P. do Carmo, *Riemannian Geometry*